Cybersecurity Course Reflection

This course helped me connect security concepts to hands-on work in our Harborstone banking lab. These are the takeaways, technical interests, and more complex ideas I would highlight in my GitHub portfolio.

Top 3 Course Quotes

These three quotes stood out to me because they connect directly to what I practiced during the course.

1. “Never trust, always verify.”

This reminded me to test whether a security control actually works. In our Windows Server lab, I checked firewall settings, account status, and audit records instead of assuming that applying a setting was enough.

2. “Document throughout, not at the end.”

During our ransomware exercise, saving evidence and recording actions as they happened helped me keep track of what we identified, preserved, and changed. Waiting until the end would make it easier to miss important details.

3. “Cybersecurity is not just an IT or CS department responsibility; it is everyone’s responsibility.”

The tabletop exercises showed me how technical teams, business operations, and employees all contribute to security. People need to report suspicious activity, follow procedures, and communicate clearly during an incident.

5 Technical Concepts That Interested Me

1. Incident Response

Our ransomware exercise made incident response one of my biggest interests. I liked learning how to identify suspicious activity, preserve evidence, contain an account, and plan recovery while keeping the response organized.

2. Windows Server Hardening

I enjoyed hardening our Windows Server by reviewing updates, disabling the Print Spooler, configuring account lockout settings, and checking firewall rules. These tasks helped me see how individual changes reduce a system’s exposure.

3. Active Directory and Group Policy

Working with Active Directory, named administrator accounts, groups, and Group Policy interested me because these controls shape who can access the banking environment and what they can do.

4. PowerShell for Security Monitoring

I liked using PowerShell to review Windows event logs, Defender status, services, and Server Message Block (SMB) file-sharing sessions. Building a daily security check showed me how scripts can make monitoring more consistent.

5. Governance Risk and Compliance
Our discussions about bank policies and independent audits made me more interested in governance, risk, and compliance (GRC). I liked seeing how technical controls support business policies and how an independent review can identify gaps that self-reporting might miss.

7 Complex Concepts I Did Not Know Before

1. Effective Permissions
Reviewing access to the BankData share helped me understand how group membership, shared-folder permissions, and New Technology File System (NTFS) permissions combine to determine actual access. I learned why checking a single permission setting is not enough.
2. Evidence Preservation and Chain of Custody
Preserving information about the apexsvc account and the ransom note helped me understand why evidence should be collected before cleanup. I also learned that a chain of custody records who handled evidence and when.
3. Network Segmentation and Lateral Movement
The VLAN 40 tabletop scenario helped me understand how a virtual local area network (VLAN) separates traffic and how isolation can limit lateral movement. I learned why cutting off one suspicious device may be preferable to disrupting an entire business segment.
4. Ransomware Recovery and Validation
The ransomware exercise helped me understand the difference between containment, eradication, and recovery. Disabling an account limits access, removing persistence addresses ways an attacker could return, and recovery requires restoring and validating normal service.
5. Proving Data Exfiltration
When I checked SMB sessions and open files, I learned that no output only describes what that check found at that moment. It does not prove that data was never stolen; confirming past exfiltration requires historical evidence such as network logs or transfer records.
6. Security Baselines and File Integrity Monitoring
The missing BankData hash baseline in my security check showed me why a trusted starting point matters. File hashes help detect content changes, while a saved file inventory helps identify missing files.
7. CVE and CVSS in Vulnerability Prioritization
Our vulnerability work helped me separate Common Vulnerabilities and Exposures (CVE), which identifies a known vulnerability, from the Common Vulnerability Scoring System (CVSS), which rates severity. I learned that deciding what to fix first also requires considering exposure, exploitation, and business impact.